

flspcell_fast_1p32V_m40C Library

Cell Groups
DFRBPQ

DFRBPQ



*flspcell_fast_1p32V_m40C Cell Library: Process
flspcell_fast_1p32V_m40C, Voltage 1.32, Temp -40.00*

Truth Table

INPUT			OUTPUT
CLK	D	RESET_B	Q
R	0	1	0
R	1	1	1
x	x	0	0
x	x	1	IQ

Footprint

Cell Name	Area
sg13g2_dfrbpq1arx_1	48.98880

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	RESET_B	Q
sg13g2_dfrbpq1arx_1	0.02241	0.01536	0.02801	0.30000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_dfrbpq1arx_1	5342.10000	5956.26000	6468.96000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RR)	0.01860	0.00100	0.25342	0.32940	0.06480	0.44328	2.50740	0.30000	1.03075

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RF)	0.01860	0.00100	0.27367	0.32940	0.06480	0.46194	2.50740	0.30000	0.95718
	RESET_B->Q (FF)	0.01860	0.00100	0.31944	0.32940	0.06480	0.54026	2.50740	0.30000	1.14508

Constraint Information

Constraints(ns) for CLK rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.07980	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for CLK falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.08301	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.03668	1.26300	1.26300	-0.13222	2.50740	2.50740	-0.17119
	setup	CLK (R)	0.01860	0.01860	0.31054	1.26300	1.26300	0.36158	2.50740	2.50740	0.36894

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.01467	1.26300	1.26300	-0.11063	2.50740	2.50740	-0.16824
	setup	CLK (R)	0.01860	0.01860	0.33010	1.26300	1.26300	0.39666	2.50740	2.50740	0.43978

Constraints(ns) for RESET_B rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	recovery	CLK (R)	0.01860	0.01860	0.07825	1.26300	1.26300	0.19698	2.50740	2.50740	0.28630
	removal	CLK (R)	0.01860	0.01860	-0.07091	1.26300	1.26300	-0.19158	2.50740	2.50740	-0.28335

Constraints(ns) for RESET_B falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	RESET_B ()	0.01860	0.00000	0.07339	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.17602	0.32940	0.06480	0.18084	2.50740	0.30000	0.26597

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.19528	0.32940	0.06480	0.20439	2.50740	0.30000	0.28802
	RESET_B	0.01860	0.00100	0.18209	0.32940	0.06480	0.18460	2.50740	0.30000	0.22101

Passive power(pJ) for CLK rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04166	0.32940	0.04734	2.50740	0.12729

Passive power(pJ) for CLK falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.07683	0.32940	0.08335	2.50740	0.16547

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.02930	0.32940	0.03517	2.50740	0.11553
	(D * !RESET_B * * !Q)	0.01860	0.04166	0.32940	0.04734	2.50740	0.12729
	(!D * RESET_B * !Q)	0.01860	0.04328	0.32940	0.04896	2.50740	0.12926
	(!D * !RESET_B * * !Q)	0.01860	0.04990	0.32940	0.05555	2.50740	0.13550

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.07372	0.32940	0.08023	2.50740	0.16301
	(D * RESET_B * !Q)	0.01860	0.07683	0.32940	0.08335	2.50740	0.16547
	(D * !RESET_B * * !Q)	0.01860	0.03665	0.32940	0.04324	2.50740	0.12277
	(!D * RESET_B * Q)	0.01860	0.23036	0.32940	0.23026	2.50740	0.30960
	(!D * RESET_B * !Q)	0.01860	0.03220	0.32940	0.03880	2.50740	0.11832
	(!D * !RESET_B * * !Q)	0.01860	0.02861	0.32940	0.03514	2.50740	0.11466

Passive power(pJ) for D rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.06831	0.32940	0.06909	2.50740	0.09157

Passive power(pJ) for D falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.06772	0.32940	0.06951	2.50740	0.10290

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.06537	0.32940	0.06708	2.50740	0.09986
	(!CLK * RESET_B)	0.01860	0.11134	0.32940	0.11293	2.50740	0.14684
	(!CLK * !RESET_B)	0.01860	0.06831	0.32940	0.06909	2.50740	0.09157

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.06772	0.32940	0.06951	2.50740	0.10290
	(!CLK * RESET_B)	0.01860	0.08946	0.32940	0.09106	2.50740	0.12623
	(!CLK * !RESET_B)	0.01860	0.05248	0.32940	0.05335	2.50740	0.07649

Passive power(pJ) for RESET_B rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.01248	0.32940	0.01337	2.50740	0.04204

Passive power(pJ) for RESET_B falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.03119	0.32940	0.03132	2.50740	0.07585

Passive power(pJ) for RESET_B rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.01248	0.32940	0.01337	2.50740	0.04204
	(CLK * !D * !Q)	0.01860	0.00522	0.32940	0.00522	2.50740	0.00526
	(!CLK * D * !Q)	0.01860	0.05538	0.32940	0.05596	2.50740	0.09834
	(!CLK * !D * !Q)	0.01860	0.00926	0.32940	0.00923	2.50740	0.00928

Passive power(pJ) for RESET_B falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.21402	0.32940	0.21647	2.50740	0.29822
	(CLK * !D * !Q)	0.01860	-0.00522	0.32940	-0.00522	2.50740	-0.00526
	(!CLK * D * !Q)	0.01860	0.03119	0.32940	0.03132	2.50740	0.07585
	(!CLK * !D * !Q)	0.01860	-0.00926	0.32940	-0.00923	2.50740	-0.00928